WMSU

TAVIDM: TRAFFIC VIOLATION DETECTION AND MONITORING SYSTEM USING COMPUTER VISION AND DEEP LEARNING

Proponents:
Pending owner confirmation

Thesis Adviser:
Pending owner confirmation

WORKING DRAFT • OFFICIAL DETAILS PENDING APPROVAL



INTRODUCTION

Traffic footage takes time to review and organize.

TAVIDM uses computer vision to detect road users and track their movement across video frames.

Configured road zones and rules help flag possible events and collect evidence for review.

Authorized users assess each candidate and manage records and reports.

Statement of the Problem

- How can traffic videos be processed to identify vehicles, track their movement, and flag possible violations using configured road zones and rules?
- How can evidence, human review, violation records, and reports be managed in one web application?

Objectives

Draft general objective: develop a web-based traffic monitoring system that processes traffic videos, flags possible violations, and supports evidence review and reporting.

Specific Objectives

- Detect and track vehicles in traffic videos using YOLOv8m and ByteTrack.
- Apply configured road zones and traffic rules to flag possible violations.
- Provide evidence review and record management for authorized users.
- Generate reports and evaluate system performance using labeled data and defined test scenarios.

Results and Discussion

Dataset Collection and Preparation

- The provisional V9 derivative contains 6,280 train, 234 validation and 224 test images.
- The manifest records 399 polygon rows converted to boxes and 0 exact duplicate images.
- The derivative remains provisional; source provenance and weak-class quality checks are open.

Results and Discussion

Vehicle Detection and Tracking

- YOLOv8m produces object labels, boxes and detection confidence scores.
- ByteTrack associates observations across frames for short-term motion history.
- Track IDs are temporary and can be interrupted by occlusion or changes in view.

Results and Discussion

Road Annotation and Traffic Rules

- Six zone types and structured road annotations add location and direction context.
- The rule engine combines detections with geometry, movement, time and configured conditions.
- Missing or ambiguous prerequisites should suppress an unsupported conclusion.

Results and Discussion

Violation Candidates and Evidence

- A rule match creates a candidate for human assessment; it does not establish guilt.
- Available evidence can include a scene image, time, track information and a video record.
- Detector confidence and rule evidence strength describe different things.

Results and Discussion

Human Review and Vehicle Grouping

- Reviewers can confirm a proposed type, choose another canonical type, dismiss it or mark evidence insufficient.
- Grouping checks video, processing run, track and saved evidence identity.
- No live review or correction-and-fusion flow was demonstrated for this deck.

Results and Discussion

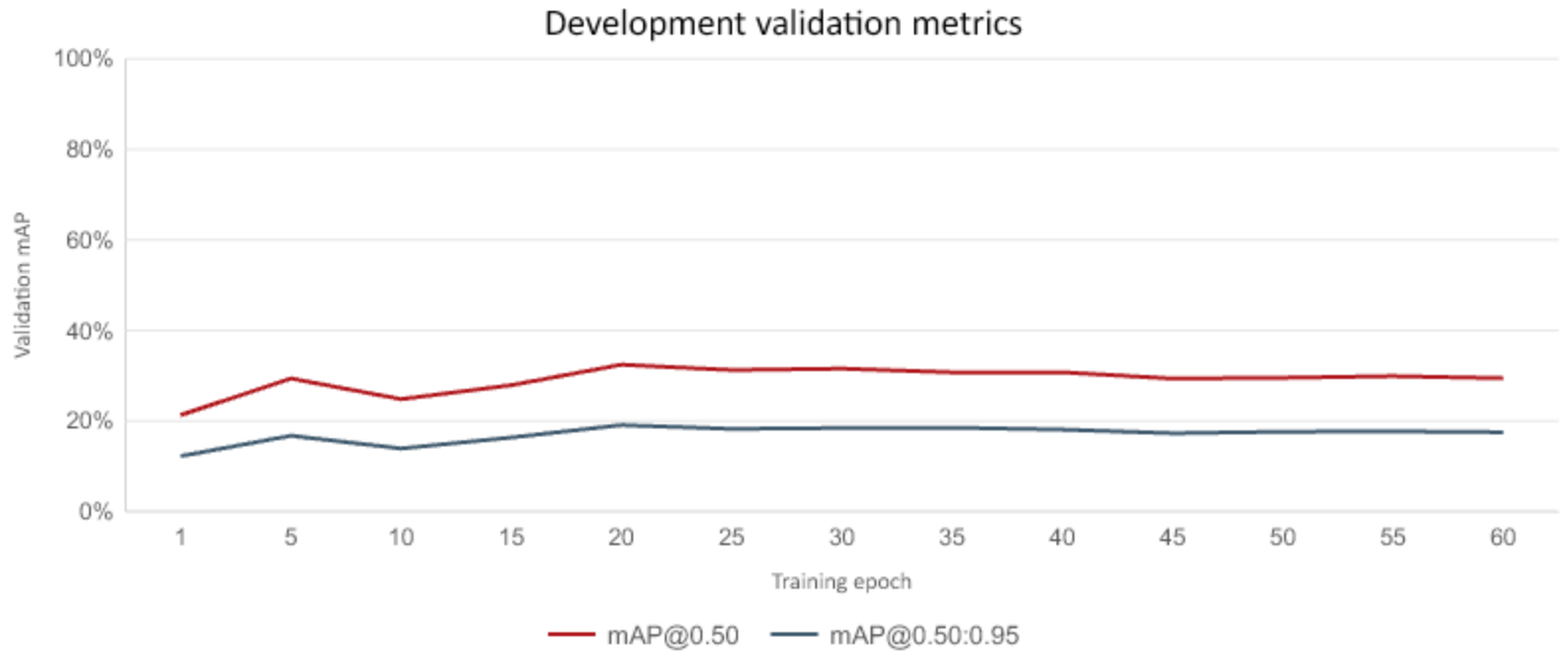
Dataset Splitting and Evaluation Method

- The provisional derivative lists 6,280 train, 234 validation and 224 test images.
- The located pilot log reports validation metrics during training; no held-out test metrics were found.
- The manifest marks training and promotion as not authorized.

Results and Discussion

Evaluation Results and Limitations

- Epoch 60: precision 38.7%, recall 34.4%, mAP@0.50 29.4%, mAP@0.50:0.95 17.5%.
- Validation only; this is not held-out test performance or violation accuracy.



Results and Discussion

System Integration and Reports

- Flask connects video input, processing, SQLite records, evidence review, analytics and PDF/Excel report routes.
- YOLOv8m, ByteTrack and the rule engine supply candidates to the review workflow.
- Source inspection does not prove current online or live-demo readiness.



Conclusions

- The source implements a video-to-candidate workflow with detection, tracking, configured rules and human review.
- The canonical roster has 12 types: 4 marked implemented and 8 partial.
- The available pilot is validation on a provisional derivative; held-out test performance remains unverified.

Recommendations

- Confirm the approved title, names, problem statement and objectives before presenting.
- Complete provenance and label review, then evaluate on a genuinely unseen authorized test set.
- Recheck partial rules and event correction/fusion with documented scenarios.
- Prepare permitted system screenshots and rehearse the authorized review and report path.
- List violations by video, grouped by category, with filters.

Demo

Live demonstration not included in this draft.

Before the defense, prepare an authorized demo account, approved video, review example and PDF or Excel report.

No application session was available to verify this flow.